INTERNAL PLATFORM REFERENCE

# MatrixIQ Feature Reference

Every module currently shipping in MatrixIQ — what it does, who can reach it, and how the pieces connect. One record follows a customer from a scanned business card to a paid invoice; the supporting modules hang off that spine.

| 52 | 10 | 15 | 11 | 86 | 310 | 1,505 |
|---|---|---|---|---|---|---|
| MODULES | SECTIONS | ROLES | DEPARTMENTS | TABLES | API ROUTES | TESTS |

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## How it fits together

MatrixIQ is not a set of independent tools. One commercial record moves through the whole system, gaining documents and approvals as it goes — and every supporting module reads from or writes to that same record rather than keeping its own copy.

`Lead` → `Project` → `Demo` → `Quotation` → `Negotiation` → `Commit` → `Delivery Challan` → `Billing` → `Payment`

> **The sales pipeline is five forward stages** — `prospect → demo → quotation → negotiation → commit` — plus the terminal `closed_lost`. A lead that qualifies becomes a project automatically, assigned to the rep who owned the lead, and waits for them to confirm it. Everything downstream hangs off that project id.

### What branches off the spine

| | |
|---|---|
| **Technical delivery**<br>`TMS` | A won project hands over to the Technical Project Management side — tasks, bill of materials, procurement — without re-keying the customer. |
| **After-sales**<br>`Support` | Support tickets attach to the project and the quotation they came from, with SLA tracking on response and resolution. |
| **Channel**<br>`Partner Portal` | External partners submit opportunities into the same pipeline. A partner project *is* a project row, flagged with a partner id — not a parallel table. |
| **People**<br>`HR & Workspace` | Daily reports, reimbursements, travel and task planning run alongside, scoped by the same department model the pipeline uses. |

## Sales `8 modules`

The commercial core — from first contact to a billable won deal.

### Lead Capture / Inquiry

`leads`

Scan a business card at an event, or bulk-import from CSV or multiple photos, and qualify each lead on the spot.

Facebook & Instagram Lead Ads feed straight in via the Meta integration. A qualified lead auto-creates its project.

- Scan a business card — OCR fills the lead
- Preview and correct before committing an import
- Log a call against a lead
- Qualify and convert to a project
- Lead statistics and Meta ad analytics
- Bulk import from CSV or a batch of photos
- Download a CSV template
- Assign or reassign to a rep
- Export the list to CSV

`capture → qualify → assign → call → convert`

### Project Dashboard

`projects`

Every sales project — site visit to close — with a full pipeline timeline.

Carries client details, departments, expected value and closing date, attachments, notes, and a stage-by-stage timeline of who changed what.

- Create, edit and close a project
- Confirm an auto-assigned project
- Change the sales person
- Raise a technical request and respond to it
- Review a partner submission — accept, decline, ask for more
- Move through the pipeline stage by stage
- Set or override the closing probability
- Request an extension to a deadline, and decide on one
- Hand a project over, cancel or respond to the handover
- Per-project KPIs and a full timeline

`prospect → demo → quotation → negotiation → commit`

### Site Visit Report

`site-visits`

Register a visit and keep logging project updates over time.

- Register a visit against a project
- Keep logging updates over time
- Upload site photographs
- Browse visits by project or status

### Demo Schedule

`demo-schedule`

Request and approve product demos.

A full approval chain: technical approval → manager approval → back office → delivery challan → demo → return.

- Raise a demo request with products and a date
- Manager approval step
- Track through to return of equipment
- Technical approval step
- Hands off to Back Office for a delivery challan

`requested → technical → manager → back office → demo → return`

### Quotations

`quotation`

Build a quotation, revise it, and track every version.

Three tabs: New Quotation, Existing Quotations, and Partner Requests. Line items priced from the product catalog; PDF export with company header and terms; statuses `draft → sent → approved / rejected` with a validity window.

- Build a quotation from the product catalog
- Revise an existing quotation into a new version
- Move a quotation's project to the demo stage
- Change status, with a validity window
- Export to PDF with company header and terms
- Apply markup, discount, GST and freight
- Compare versions side by side
- Log a follow-up against a quotation
- Share with a channel partner
- Prepare on behalf of another rep

`draft → sent → approved / rejected`

### Partner Quotation Requests

`quotation-requests`

Partners asking us to price something — prepare a quotation or decline with a reason.

Marking a request quoted shares the quotation with the partner in the same action.

- See what partners have asked us to price
- Pick which quotation answers the request
- Start preparing, so the partner sees progress
- Sending it shares the quotation with the partner
- Decline with a reason the partner reads

`open → in progress → quoted / declined`

### Billing

`billing`

Won deals and approved quotations — what is billable, what has a purchase order, and what nobody has priced yet.

- See won deals and approved quotations in one queue
- Surface deals nobody has priced yet
- Separate what is billable from what has a PO

### Partner Management

`partners`

Channel partners, distributors and dealers — accounts, access, assigned relationship manager, and everything they have submitted.

- Create a partner company with a partner code
- Assign a relationship manager
- See everything that partner has submitted
- Provision portal user accounts for them
- Activate, suspend or close an account

## Technical Project Management `7 modules`

Delivery after the deal is won, with its own roles and per-tab permission matrix.

### Technical Projects
`tms-projects`

Every technical project — status, team and progress. Open a project for its tasks, BOM requests, procurement and activity.

- Create technical projects and set phases
- Track status and progress
- Assign a team
- Open a project for its tasks, BOM, procurement and activity

### TMS Dashboard
`tms-dashboard`

Project, task, BOM, and procurement overview for the Technical Team.

- Project, task, BOM and procurement overview
- Workload across the technical team

### Tasks
`tms-tasks`

Day-by-day task tracking with a Daily Task View.

Includes deadline extension requests with an approval step.

- Create and assign tasks
- Request a deadline extension
- Daily Task View, day by day
- Approve or reject an extension request

### BOM Request
`tms-bom-requests`

Bill of materials requests, review, and approval.

Finance approval is a configurable step — see Application Settings.

- Raise a bill of materials request
- Technical approve or reject
- Admin approve or reject
- Mark payment made
- Submit for review
- Finance approve or reject
- Send to procurement
- Mark materials received

`draft → submitted → approved → finance → procurement → received`

### Procurement
`tms-procurement`

Purchase and delivery tracking from approved BOM requests.

- Track purchases from approved BOM requests
- Record delivery against each line

### Users & Tab Access
`tms-users`

Manage technical team accounts, and which TMS roles can view, create, edit, delete, approve, or manage each tab.

- Manage technical team accounts
- Set which TMS roles can view, create, edit, delete, approve or manage each tab

### Tab Access
`tms-tab-access`

Configure which TMS roles can view, create, edit, delete, approve, or manage each TMS tab.

- The per-tab permission matrix on its own screen

## Workspace `4 modules`

What every employee uses daily, whatever department they sit in.

### My Tasks

`my-tasks`

Tasks assigned to you, from any module.

- Every task assigned to you, from any module
- Request a deadline extension
- Update progress and mark complete

### Team Tasks

`team-tasks`

Assign and track tasks within your own department.

- Assign work within your department
- Reassign to someone else
- Review submitted work
- Track against deadlines

### Task Planner

`admin-task-assignment`

Plan, assign and track work across departments and employees.

- Plan and assign across departments
- Track to completion
- See load per employee

### Daily Report

`daily-report`

File your daily activity report, claim leave for a day you were away, and — if you manage a team — see who has filed.

Each department gets its own report format. Managers can now edit their department's task categories and add a sub-category level without a code change.

- File today's report on your department's format
- Claim leave for a day you were away
- Request an edit after submitting, and track the decision
- Add as many task rows as the day needed
- See your own submission history
- Managers: switch to the team view

`draft → submitted (on time / late) → reviewed`

> **Daily reports are deadline-driven.** A configurable working-day calendar and cut-off time decide what counts as late; a scheduler sends evening and morning reminders, and raises a warning for a missed report. A day covered by an approved leave claim is never counted as missing.

## HR & Admin `7 modules`

People, spend, and the approval chains around both.

### Employees
`hr-employees`

Active employee directory.

Includes the HR dashboard with birthdays and work anniversaries.

- Active employee directory
- Birthdays and work anniversaries
- Employee exit checklist, summary and commit
- Open one employee's full profile
- Attendance view

### Office Operation Expenses
`office-operation-expenses`

HR/Admin office operating spend — office, electricity, guest, director, salary, and pantry expenses.

Category and Expense Head dropdowns are now editable in-app, both with sub-levels.

- Record office spend with category, expense head and sub-items
- Attach the bill
- Export the register to Excel
- Pick the office location and who paid
- Edit the Category and Expense Head dropdowns in-app, with sub-levels

### Reimbursement
`reimbursement`

Submit and track expense reimbursement bills.

Multi-stage workflow with email notification at each stage, and Excel voucher export.

- Submit bills into a monthly sheet
- Track which bills are still missing
- Manager decision
- Accounts completion
- Replace a bill without losing the entry
- Submit the sheet for approval
- HR decision
- Download the voucher as Excel

`draft → submitted → manager → HR → accounts → paid`

### Admin Expenses
`admin-expenses`

Add hotel & ticket expenses split across employees.

- Add hotel and ticket costs
- Split one expense across several employees

### HR Expense Report
`hr-reports`

Reimbursement, admin expense, and office operation spend by period, employee, or department.

- Reimbursement, admin and office spend in one report
- Export the result
- Filter by period, employee or department

### Travel Schedule
`travel-schedule`

Log rep travel for client visits.

Four-stage approval: manager → HR → Admin → Accounts, with ticket booking in between.

- Log planned travel for a client visit
- Manager decision
- Admin decision
- HR final verification
- Submit for approval
- HR review
- Complete the ticket booking

`submitted → manager → HR → admin → booking → verified`

### Daily Report Review
`hr-daily-reports`

Approve or reject leave claims, and review or waive missed-report warnings.

- Approve or reject a leave claim
- Waive a warning with a reason
- Review missed-report warnings

## Marketing 2 modules

### Marketing Requests

`marketing-requests`

Request marketing support — brochures, banners, social posts, and more — and track delivery timelines.

A seven-step workflow: submit → assign → accept → marketing work → technical review → final submission → deliver. Requests raised by channel partners are badged as such in the queue.

- Raise a request through a four-step wizard
- Assign to a marketing member
- Set a delivery timeline
- Send to technical review
- Deliver the final files to the requester
- Cancel or reject with a reason
- Attach reference files
- Accept or decline the assignment
- Change priority
- Record the technical decision
- Comment thread on every request

```
submitted → assigned → accepted → in progress → technical review → delivered
```

### Materials

`materials`

Datasheets, case studies, brochures, presentations and certificates — the shared marketing library.

Folder tree, per-product grouping, and a per-document "Share with partners" toggle that publishes it to the Partner Portal.

- Add a document by Drive link
- Group by category and product
- Search the whole library
- Organise into a folder tree
- Toggle 'Share with partners' per document and per folder

## Operations & Support 2 modules

### Back Office Operations

`backoffice`

Delivery Challans — prepare, dispatch, verify returns, close.

Drives demo equipment in and out, and produces the monthly summary run.

- Prepare a delivery challan
- Verify the return
- Monthly summary run
- Dispatch and record what went out
- Close the challan

```
prepared → dispatched → returned → verified → closed
```

### Technical Support

`support-tickets`

After-sales support tickets — triage, assignment, SLA tracking, and resolution.

SLA policies set response and resolution targets by priority; breach is computed at read time, never stored stale.

- Raise a ticket against a project or quotation
- Assign to an engineer
- SLA response and resolution targets, breach computed live
- Triage and set priority
- Comment thread with the reporter
- Resolve and close

## Accounts 2 modules

### Payments
`accounts-payments`

Every pending, on-hold, and completed payment across every module — the Accounts team's single queue.

- Every pending payment across every module in one queue
- Put on hold and resume
- Export to CSV
- Mark as paid
- Payment history and summary
- Reimbursement report

### Audit Log
`audit-log`

Every status-changing action across the Back Office workflow.

Records who, what, previous and new status, remarks and IP — written from every module that changes a record's state.

- Who did what, when, and from which IP
- Filter by entity, person or period
- Previous and new status on every change

## Reports 5 modules

Reporting reads the same records the operational modules write — there is no separate warehouse to fall out of step.

### Analytics
`analytics`

Quotation, project, and pipeline performance at a glance.

- Quotation, project and pipeline performance
- Trends over a chosen period

### Target Details
`targets`

Sales targets vs achievement, by employee and period.

- Set sales targets by employee and period
- Weekly update entry
- Track achievement against target
- Drill into one employee's numbers

### Performance Review
`performance-review`

A full performance dashboard for one employee at a time — CRM, sales, projects, and activity history.

- One employee's CRM, sales, project and activity history
- A full period-over-period picture

### Team Daily Reports
`team-daily-reports`

Who filed, who filed late, and who is missing — by date and department. Review a report, or file one on an employee's behalf.

Two views: a Day view for chasing today's deadline, and a Sheet view — a tab per employee with a flat day-by-day task sheet, plus an all-employees status grid and CSV export.

- Day view — who filed, who was late, who is missing
- All-employees status grid across a date range
- File a report on an employee's behalf
- Export to CSV
- Sheet view — a tab per employee, day-by-day tasks
- Open and review any report
- Edit a department's task categories

### Partner Analytics
`partner-analytics`

Compare partner performance — submissions, conversion, demos and won business.

- Compare partners on submissions and conversion
- Demos requested and business won

## Partner Portal 7 modules external

A separate experience for channel partners, distributors and dealers — running on the same

database, the same tables and the same deployment as the internal app.

### Dashboard
`partner-dashboard`

Your submissions, demos, quotations and support at a glance.

- Submissions, demos, quotations and support at a glance
- What is waiting on you

### My Leads / Projects
`partner-projects`

Submit a new opportunity and track every one you have sent us.

Same form fields as the internal New Project screen, minus the internal-person pickers. When Sales asks for changes, the partner can edit the submission and every change is logged by name.

- Submit a new opportunity on the full project form
- Track it through the pipeline
- Edit the submission while changes are requested
- Attach supporting files
- Reply when Sales asks for more information
- See the whole Sales conversation in one thread

`submitted → under review → info requested → accepted / declined`

### Quotations
`partner-quotations`

Quotations shared with you, with their versions and validity.

Read and download only — a partner can never revise a quotation.

- Every quotation shared with you
- Download the PDF
- Version history and validity

### Demo Requests
`partner-demos`

Request a product demonstration for your customer and follow it to completion.

The technical person is chosen by the assigned salesperson, never by the partner.

- Request a demo for your customer
- Follow it through to completion
- State the date and requirements you need

### Marketing Materials
`partner-materials`

Brochures, datasheets, case studies and presentations published for partners.

- Browse published brochures, datasheets and case studies
- Search the published library
- Folder tree with category and product grouping
- Open or download any document

### Marketing Requests
`partner-marketing-requests`

Ask our marketing team for collateral and track it to delivery.

- Ask for collateral on the same form our team uses
- Track progress through to delivery
- Set priority and a needed-by date
- Download the finished files

`received → in progress → in review → completed`

### Technical Support
`partner-support`

Raise and track after-sales support tickets.

- Raise an after-sales ticket
- Track response and resolution
- Describe product, serial, location and priority
- Comment back and forth with the engineer

**Three independent isolation layers, not one.** `proxy.ts` blocks the route area in both directions — a partner cannot leave `/partner/*`, and staff cannot enter it. `getViewerContext()` returns null for any partner account unless a route explicitly opts in, so every internal API refuses them by default. And `partner_id` is mandatory in every partner-facing query. Partner-facing payloads are built from an *allow-list* of fields, never by deleting sensitive ones from an internal record.

# Administration 11 modules

Configuration that changes how the rest of the platform behaves — without a deploy.

### User Management
`user-management`

Create and manage login accounts, roles, and access.

- Create and edit login accounts
- Bulk import users — preview, then commit
- See one user's activity history
- Assign role and department
- Resend the welcome e-mail

### Role Management
`role-management`

What each role can see and do across the platform.

- Define what each role can see and do
- Set the privileged flag
- Clone an existing role as a starting point

### Department Master
`department-master`

Departments used across user profiles.

Also the source of department managers, which drives visibility scoping everywhere.

- Create and rename departments
- Reorder the list
- Appoint department managers

### Module Manager
`module-manager`

Enable, disable, rename, and reorder every module.

Disabling a tile closes its APIs too — the nav and the backend cannot drift apart.

- Enable or disable any module
- Set which roles see each module
- Rename and reorder tiles
- Disabling a tile closes its APIs too

### Custom Module Builder
`custom-modules`

Create new business modules without writing code.

- Define a new module and its fields
- Export its records to CSV
- Capture records against it

### Products
`product-master`

Manage quotation products, and rename or reprice any AV, Robotics, AI Analytics & VisitIQ catalog item.

- Add and edit quotation products
- Bulk reprice
- Duplicate an existing product
- Import and export as CSV or Excel

### Product Catalog
`product-catalog-overrides`

Rename or reprice any AV, Robotics, AI Analytics & VisitIQ product used in quotations.

- Rename or reprice any catalog item
- Overrides sit on top of the base catalog

### Application Settings
`app-settings`

Company details, tax, terms, and numbering.

Feeds the quotation PDF header and the document numbering series.

- Company details, GST and PAN
- Tax rate, currency and terms
- Bank details for documents
- Document numbering series

### Meta Lead Integration
`meta-lead-integration`

Connect Facebook & Instagram Lead Ads to Lead Capture / Inquiry.

- Connect Facebook and Instagram Lead Ads
- Sync leads on demand
- Test the connection

### Daily Report Configuration
`daily-report-config`

Department report formats, working days, deadlines, reminders, warning thresholds, and the reminder scheduler.

- Assign a report format per department
- Configure reminder and warning thresholds
- Run the sweep manually
- Set working days and the deadline time
- Exempt specific employees

### Audit Log
`audit-log`

Platform-wide record of every status-changing action.

## Access model

Who sees what is decided by four things working together: the role, the *privileged* flag on that role, the department the user belongs to, and Module Manager's per-module visibility.

| ROLE | KEY | PRIVILEGED | TYPICAL REACH |
|---|---|---|---|
| Super Admin | `superadmin` | yes | Everything, including configuration |
| Admin | `admin` | yes | Everything operational, plus most configuration |
| Manager | `manager` | yes | Org-wide records; approvals for their chain |
| Technical Team | `technical` | yes | TMS plus the technical side of sales records |
| Sales | `user` | no | Their own leads, projects and quotations |
| Technical Manager | `technical-manager` | no | Their department's team and work |
| Team Lead / Engineer / Technician | `team-lead` · `engineer` · `technician` | no | Assigned TMS work |
| HR | `hr` | no | People modules; reads every department's daily reports, manages its own |
| Accounts | `accounts` | no | The payments queue |
| Back Office | `backoffice` | no | Delivery challans and the demo pipeline |
| Marketing | `marketing` | no | Marketing requests and the materials library |
| Partner Admin / Partner User | `partner-admin` · `partner-user` | no `external` | The Partner Portal only — never an internal route |

**Read and act are not the same permission.** HR reads every department's daily reports but can only edit and file within its own — widening visibility deliberately did not widen authority. The same split applies to department managers elsewhere.

## Platform notes

| AREA | DETAIL |
|---|---|
| Stack | Next.js (App Router) · React · TypeScript · Sequelize · PostgreSQL · CSS Modules |
| Scale | 86 database models · 122 migrations · 310 API routes · 85 pages · 120 components |
| Tests | 1,505 unit tests across 82 files, concentrated on access rules, workflow state machines and pure business logic |
| Sessions | HMAC-signed token in an httpOnly cookie, carrying role, privileged flag, and partner id |
| Notifications | One in-app ledger shared by every module, with per-audience link resolution; e-mail on the lifecycle events that warrant it |
| Documents | Quotation and proposal PDFs, Excel exports for expenses, vouchers, daily report sheets and the payments queue |
| Scheduling | Daily report reminders and sweeps run on a scheduler; SLA and deadline states are computed at read time rather than stored |

**Soft delete is the default.** Most tables are paranoid — a deleted record disappears from every list and report but remains recoverable. Audit entries and the notification ledger are append-only.

Generated from the live module registry and database schema. Module keys match `lib/moduleConfigStore.ts`; counts reflect the current working tree.